Supplementary Material for the paper "LUNAR: Automated Testing of Multi-Turn Conversations in Task-Oriented LLM Systems"

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Table 1: System intents and their descriptions.

Intent	Description
`inform`	The system provides information relevant to the user's query without a follow-up question or call to action.
`inform_and_followup`	The system provides information and additionally poses a follow-up question or proposes an action.
`choice`	The system offers at least two concrete options to choose from, with no follow-up question.
`choice_and_followup`	The system offers at least two options and additionally poses a follow-up question.
`confirm`	The system confirms the requested action, without a follow-up question.
`confirm_and_followup`	The system confirms the requested action and additionally poses a follow-up question.
`clarify`	The system asks the user a clarifying question to resolve ambiguity.
`reject`	The system indicates it cannot fulfill the request, without a follow-up question.
`reject_and_followup`	The system indicates it cannot fulfill the request and poses a follow-up question or suggests an alternative.
`failure`	The system signals that something went wrong (e.g. it did not understand the input).
`misc`	The system's response does not fit any other category.

Table 2: (CarControl) Overview of features with categories and corresponding ranges. Null indicates that this featues is not used in the generation.

Feature	Category	Range
politeness	style	unpleasant < neutral < polite
anthropomorphism	style	very directive < directive < interrogative < empathic
slang	style	formal < neutral < rude
implicitness	style	not implicit < slightly implicit < implicit
word_perturbation	perturbation	{null, delete_words, introduce_fillers_llm}
system	content	{windows, fog_lights, ambient_lights, head_lights, climate, fan, seat_heating}
position	content	{null, driver, passenger, back_left, back_right}
seat_position	content	{null, driver, front_passenger, back_left, back_center, back_right}
fog_light_position	content	{null, front, rear}
window_state_target	content	{open, close}
onoff_state_target	content	{on, off}
head_lights_mode_target	content	{null, off, parking, low_beam, high_beam, auto}
climate_temperature_value_target	content	{null, 16–26}
seat_heating_level_target	content	{null, 0–3}
choice	intent	[0.001, 1.0]
add_preferences	intent	[0.001, 1.0]
ask	intent	[0.001, 1.0]
change_of_mind	intent	[0.001, 1.0]
confirmation	intent	[0.001, 1.0]
reject	intent	[0.001, 0.3]
reject_clarify	intent	[0.001, 0.5]
repeat	intent	[0.001, 0.5]
stop	intent	[0.001, 0.1]

Table 3: (Navi) Overview of features with categories and corresponding ranges.

Feature	Category	Range
category	content	{car_repair, restaurant, supermarket, bakery, bar, hotel, cinema, gas_station, hospital}
payment_method	content	{null, CASH, CREDIT_CARD, CONTACTLESS}
fuel_type	content	{null, SP95_E10, DIESEL, SP95, SP98}
food_type	content	{null, german, indian, italian, middle_eastern, french, chinese, japanese, thai, mexican, greek, vietnamese, turkish, american, vegan}
parking	content	{null, available}
word_perturbation	perturbation	{null, delete_words, introduce_fillers_llm, introduce_homophones_static}
rating	content	{null, 3.5, 4, 4.5}
fuel_price	content	{null, 1.5–1.9}
slang	style	formal < neutral < rude
implicitness	style	not implicit < slightly implicit < implicit
politeness	style	neutral < polite
anthropomorphism	style	very directive < directive < interrogative < empathic
choice	intent	[0.001, 1.0]
add_preferences	intent	[0.001, 1.0]
ask	intent	[0.001, 0.7]
change_of_mind	intent	[0.001, 1.0]
confirmation	intent	[0.99, 1.0]
reject	intent	[0.001, 0.3]
reject_clarify	intent	[0.001, 0.5]
repeat	intent	[0.001, 0.1]
stop	intent	[0.001, 0.1]

Table 4: Intent transition mapping: overview of permitted user intents which can follow system intents.

System Intent	Permitted Follow-up User Intents
`inform`	`add_preferences`, `ask`, `change_of_mind`, `stop`
`inform_and_followup`	`change_of_mind`, `add_preferences`, `confirmation`, `reject`, `reject_clarify`, `ask`, `stop`
`choice`	`choice`, `ask`, `add_preferences`, `change_of_mind`, `reject`, `reject_clarify`, `stop`
`choice_and_followup`	`choice`, `add_preferences`, `change_of_mind`, `reject`, `reject_clarify`, `stop`
`confirmation`	`change_of_mind`, `ask`, `add_preferences`, `stop`
`confirm_and_followup`	`change_of_mind`, `add_preferences`, `confirmation`, `stop`
`clarify`	`add_preferences`, `change_of_mind`, `ask`, `reject`, `stop`
`reject`	`change_of_mind`, `ask`, `stop`
`reject_and_followup`	`change_of_mind`, `confirmation`, `stop`
`failure`	`change_of_mind`, `repeat`, `stop`
`misc`	`change_of_mind`, `reject`, `reject_clarify`, `stop`

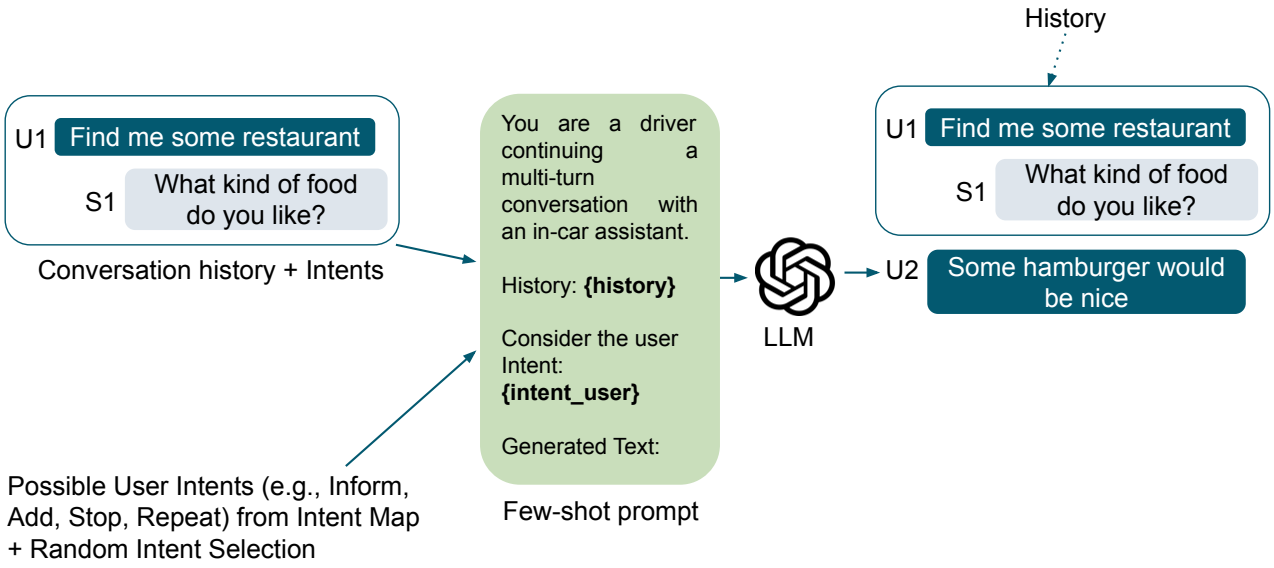

Figure 1: Example of follow-up turn simulation. LLM is instructed with conversation history and chosen intent to generate follow-up turn. Intent selection is based on intent map, classified system intent and intent priority.

Table 5: Inter-rater and judge-human agreement for judge evaluation for Navi and CarControl use case. Interrate agreement for humans is calculated using fleiss kappa, while for judge-human agreement Cohens kappa is used.

		Request-orientation		Clarity	
Case Study	Model	α	κ	α	κ
Navi	DeepSeek-V3	0.88	0.68	0.82	0.59
	GPT-4o	0.83	0.58	0.48	0.30
	GPT-4o-mini	0.68	0.48	0.23	0.05
	GPT-5-mini	**0.90**	**0.79**	0.75	0.49
	Human†	0.81	0.66	0.64	0.39
Car Control	DeepSeek-V3	0.64	0.65	0.67	0.66
	GPT-4o	0.57	0.60	0.33	0.43
	GPT-4o-mini	0.57	0.50	0.19	0.316
	GPT-5-mini	**0.65**	**0.65**	0.72	0.71
	Human†	0.76	0.57	0.61	0.35

Figure 2: Example of turn-wise effectiveness evaluation: output point of interest (POI) is compared against input requirements to assess overall effectiveness score.

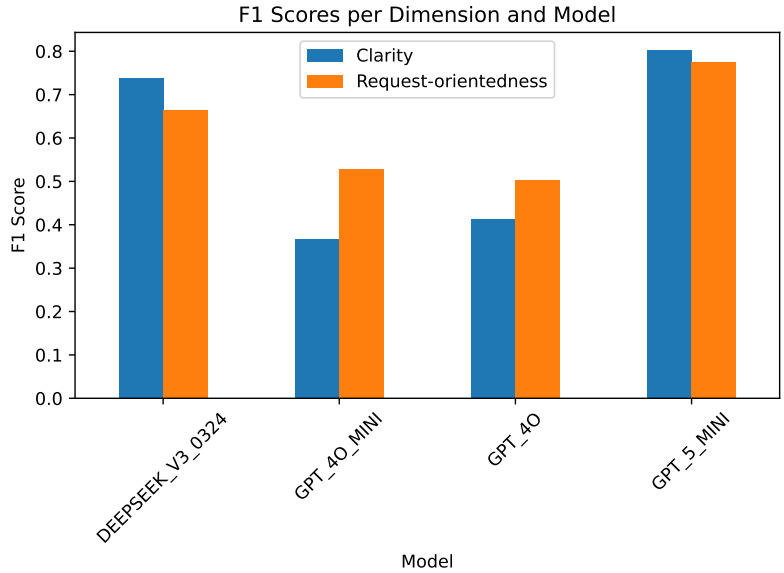

Figure 3: F1-score results for judge-based request response evaluation for car control case study for different models. Ground truth scores are computed via majority votes from human survey.

Table 6: Effectiveness (mean failures found) split by SUT model averaged over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	13.29	159.33	120.67	419.83
	GPT-5-Chat	13.50	85.56	137.00	314.57
LUNAR_S	GPT-4o	8.83	78.50	73.83	125.00
	GPT-5-Chat	14.83	24.33	114.00	84.00
SENSEI	GPT-4o	0.14	23.57	31.38	23.50
	GPT-5-Chat	0.00	7.67	11.33	17.17

Table 7: Failure rate (mean critical ratio) split by SUT model average over 6 runs.

Approach	LLM	CarControl-II	CarControl-I	Navi-II	Navi-I
LUNAR	GPT-4o	0.099	0.470	0.371	0.591
	GPT-5-Chat	0.141	0.326	0.292	0.597
LUNAR_S	GPT-4o	0.071	0.233	0.278	0.256
	GPT-5-Chat	0.171	0.124	0.278	0.329
SENSEI	GPT-4o	0.001	0.103	0.109	0.051
	GPT-5-Chat	0.000	0.042	0.047	0.051

Table 8: Efficiency results (time per failure, hours/failure) split by SUT model. Lower is better.

Approach	LLM	CarControl-II (2.0h)	CarControl-I (2.0h)	Navi-II (3.5h)	Navi-I (4.0h)
LUNAR	GPT-4o	0.1505	0.0126	0.0290	0.0095
	GPT-5-Chat	0.1481	0.0234	0.0255	0.0127
LUNAR_S	GPT-4o	0.2264	0.0255	0.0474	0.0320
	GPT-5-Chat	0.1348	0.0822	0.0307	0.0476
SENSEI	GPT-4o	14.0000	0.0849	0.1116	0.1702
	GPT-5-Chat	–	0.2609	0.3088	0.2330

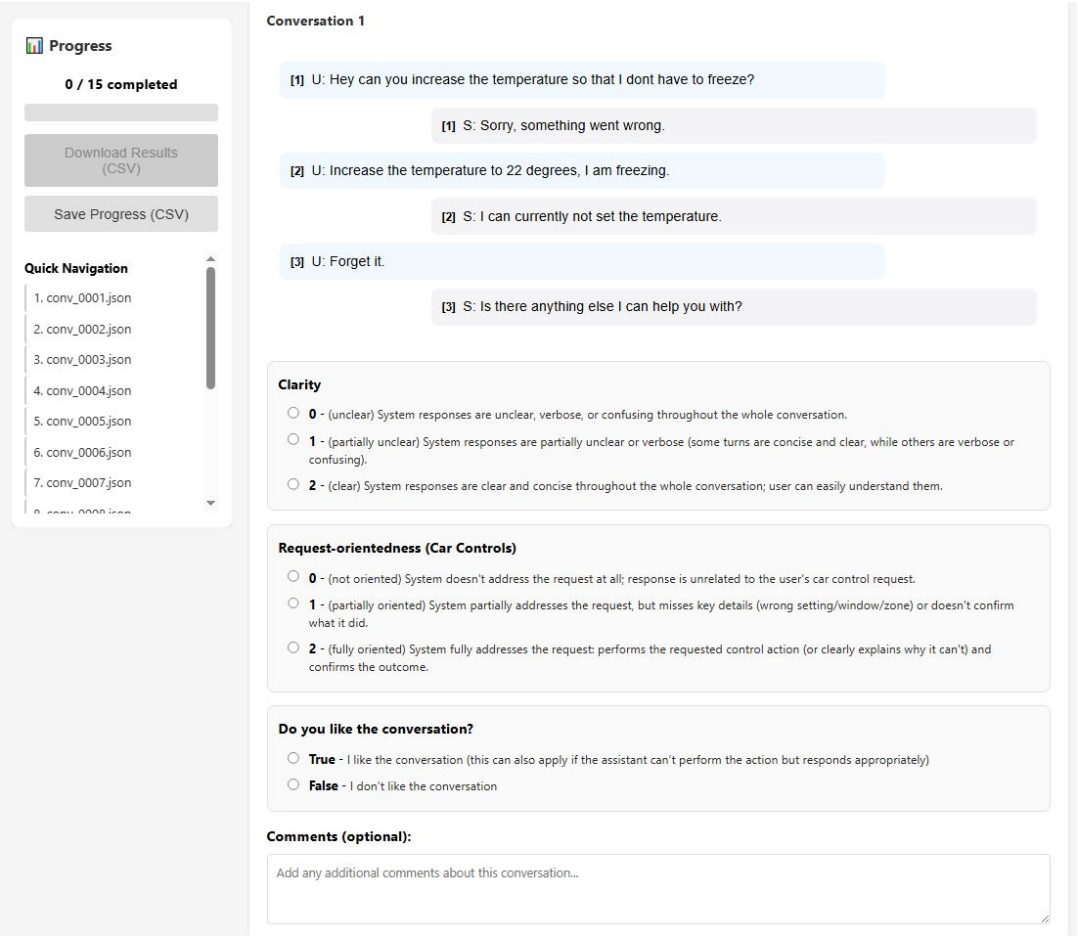

Figure 4: Overview of user interface for the survey for human-based data annotation for judge evaluation.